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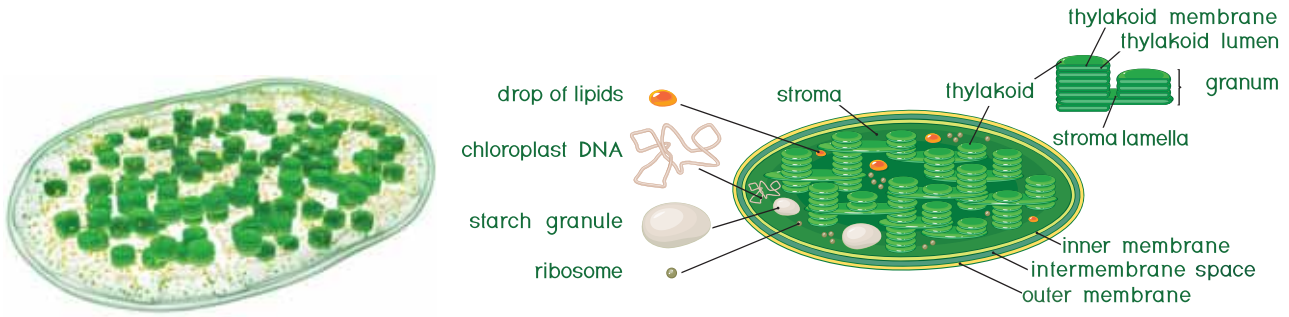


Figure 13.1: (a) 3D view of chloroplast (b) Sectional view of chloroplast

solar energy. Stroma contains enzyme which reduces carbon dioxide into carbohydrates. In Cyanobacteria thylakoid lies freely in cytoplasm without envelope (Figure 13.1).

13.3 Photosynthetic Pigments

A photosynthetic pigment is a pigment that is present in chloroplasts or photosynthetic bacteria which captures the light energy necessary for photosynthesis (Table 13.1).

13.3.1 Chlorophyll

Chlorophyll a is the primary pigment which acts as a reaction centre and all other pigments act as accessory pigments and trap solar energy and then transfer it

Table 13.1: Types of Photosynthetic pigments

Chlorophyll	Carotenoids	Phycobilins
1. Chlorophyll 'a' ($C_{55}H_{72}O_5N_4Mg$) – Green plants and Cyanobacteria	1. Carotene ($C_{40}H_{56}$) – Lycopene (Red)	1. Phycocyanin – Cyanobacteria
2. Chlorophyll 'b' ($C_{55}H_{70}O_6N_4Mg$) – Green algae and all higher plants	2. Xanthophyll ($C_{40}H_{56}O_2$)- Yellow colour – Violaxanthin, Fucoxanthin (Brown Algae) and Lutein	2. Phycoerythrin – Red Algae
3. Chlorophyll 'c' ($C_{55}H_{32}O_5N_4Mg$) – Dinoflagellates, Diatoms and Brown Algae		
4. Chlorophyll 'd' – Red Algae		
5. Chlorophyll 'e' – Xathophycean Algae		
6. Bacteriochlorophyll 'a'		
7. Bacteriochlorophyll 'b'		
8. Chlorobium Chlorophyll 650		
9. Chlorobium Chlorophyll 666		